

28/9/26

Ex.no:8 Nmap line Host Discovery

ARP, ICMP and TCP / UDP Ping Scans:

Aim:-

To discover live hosts on a network using Nmap. Host discovery scans - and to understand which protocol operate at which layer of the TCP/IP model, and when each one applies.

ARP Host Discovery:

```
$ sudo nmap -PR -sn 10.200.6.0/24
```

Op:-

Starting Nmap 7.80 (<http://nmap.org>)

Nmap scan report for 10.200.6.1

Host is up.

Nmap done: 256 IP addresses (1 host up) scanned in 10.51 seconds.

ICMP Host discovery.

7.1) ICMP Echo (-PE):

```
$ sudo nmap -PE -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.1

Host is up (0.069s latency)

Nmap scan report for 10.200.6.50

Host is up (0.070s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s latency)

Nmap done: 256 IP addresses (2 hosts up) scanned in 8.44 seconds.

ICMP Timestamp (-PP)

```
$ nmap -PP -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.50

Host is up (0.069s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s latency)

Nmap done: 256 IP addresses (2 hosts up) scanned in 8.44 seconds.

TCP Address Mask (-PM)

```
$ nmap -PM -sn 10.200.6.0/24
```

Nmap done: 256 IP addresses (0 hosts up)
Scanned in 52.14 seconds.

TCP & UDP ping Discovery

(i) TCP SYN Ping.

```
$ nmap -PS -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.50

Host is up (0.070s latency)

Nmap scan report for 10.200.6.250

Host is up (0.068s latency)

Nmap done: 256 IP addresses (2 hosts up)
Scanned in 33.88 seconds.

(ii) TCP ACK Ping (-PA)

```
$ sudo nmap -PA -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.50

Host is up (0.070s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s latency)

Nmap done: 256 IP addresses (2 hosts up)
Scanned in 33.64 seconds.

(iii) UDP Ping (-PU)

```
$ sudo nmap -PU -sn 10.200.6.0/24
```

Nmap scan report for 10.200.6.50

Host is up (0.070s latency)

Nmap scan report for 10.200.6.250

Host is up (0.069s latency)

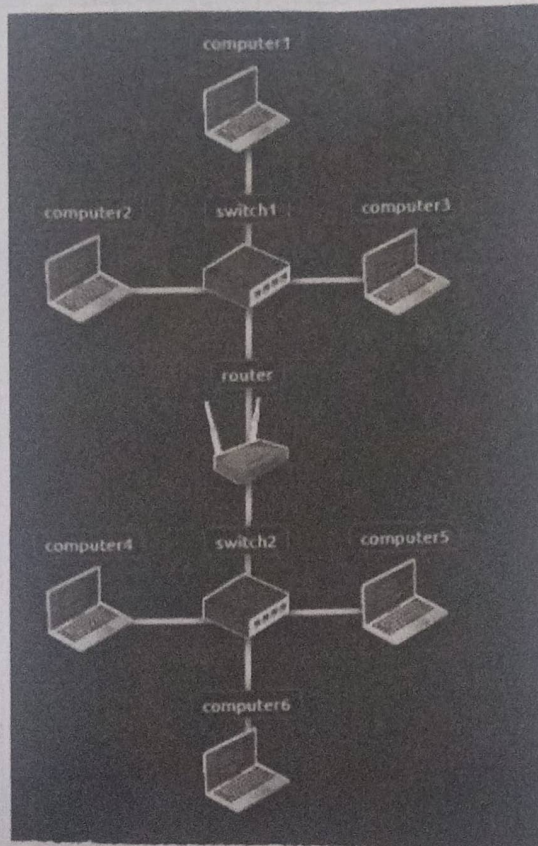
Nmap done: 256 IP addresses (2 hosts up)
Scanned in 35.97 seconds.

Marscan - A Faster Alternative

\$ marscan 10.200.6.0/24 -p 743

\$ marscan 10.200.6.0/24 -p 80,743

\$ marscan 10.200.6.0/24 -p 22-25



Answer the questions below

How many hosts are found alive after scanning the CONNECTION IP/24?

✓ Correct Answer

What is the option required to ARP request to use ICMP Echo during to discover live hosts?

✓ Correct Answer

What is the option required to ARP request to use ICMP Address Mask to discover live hosts?

✓ Correct Answer

What is the option required to ARP request to use ICMP Echo to discover live hosts?

✓ Correct Answer

Which TCP ping scan does not require a privileged account?

✓ Correct Answer

Which TCP ping scan requires a privileged account?

✓ Correct Answer

Which option do you need to add to Nmap to run a TCP SYN ping scan on the target host?

✓ Correct Answer

8/16

Result:

Thus live host discovery for ARP, ICMP, TCP, UDP was studied and the output got verified successfully.